

Laboratorio 7 - Ejercicio 1

$$\textcircled{1} \quad S \rightarrow \emptyset A \emptyset \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \varepsilon$$

a) elim. prod ε

$$C \rightarrow \varepsilon$$

$$A \rightarrow C$$

$$\hookrightarrow A \rightarrow C \rightarrow \varepsilon$$

$$B \rightarrow A$$

$$\hookrightarrow B \rightarrow A \rightarrow \varepsilon$$

$$S \rightarrow BB$$

$$\hookrightarrow S \rightarrow BB \rightarrow \varepsilon \varepsilon$$

S, A, B, C
son variables

$$S \rightarrow \emptyset A \emptyset$$

$$\hookrightarrow S \rightarrow \emptyset A \emptyset$$

$$\hookrightarrow S \rightarrow \emptyset \emptyset$$

$$S \rightarrow BB$$

$$\hookrightarrow S \rightarrow BB$$

$$S \rightarrow B \quad \vee \quad S \rightarrow B$$

$$S \rightarrow \varepsilon$$

$$\underbrace{\hspace{10em}}$$

$$\hookrightarrow S \rightarrow BB \mid B$$

$$S \rightarrow 1B1$$

$$\hookrightarrow S \rightarrow 1B1$$

$$S \rightarrow 11$$

$$S \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB \mid B$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

b) elim. prods. unitarias

$$\left. \begin{array}{ll} S \rightarrow B & B \rightarrow A \\ A \rightarrow C & C \rightarrow S \\ B \rightarrow S \end{array} \right\} \begin{array}{l} \text{prods.} \\ \text{unitarias} \end{array}$$

↳ pares = (S, S) (B, B)
 (A, A) (C, C)

$S \rightarrow B \rightarrow A \rightarrow C \rightarrow S \rightarrow \dots$ lo = para A, B, C
forman ciclo

$S \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$
 $A \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$
 $B \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$
 $C \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$

c) elim. símbolos useless

1) $S, A, B, C \rightarrow$ generadores

2) $S \rightarrow \emptyset A \emptyset \mid 1B1 \mid BB$

alcanzables = S, A, B

$S \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$
 $A \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$
 $B \rightarrow \emptyset A \emptyset \mid \emptyset \emptyset \mid 1B1 \mid 11 \mid BB$

$$(2) \quad S \rightarrow aAa \mid bBb \mid c$$

$$A \rightarrow c \mid a$$

$$B \rightarrow c \mid b$$

$$C \rightarrow CDE \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

a) elim. prods. ϵ

$$\hookrightarrow S \rightarrow \epsilon$$

$$C \rightarrow \epsilon$$

$$\hookrightarrow A \rightarrow C$$

$$\hookrightarrow A \rightarrow C \rightarrow \epsilon$$

$$\hookrightarrow B \rightarrow C$$

$$\hookrightarrow B \rightarrow C \rightarrow \epsilon$$

$$\hookrightarrow D \rightarrow A$$

$$\hookrightarrow D \rightarrow A \rightarrow \epsilon$$

S, A, B, C, D
son anulables

$$S \rightarrow aAa$$

$$\hookrightarrow S \rightarrow aAa$$

$$S \rightarrow aa$$

$$C \rightarrow CDE$$

$$\hookrightarrow C \rightarrow CDE$$

$$C \rightarrow DE$$

$$C \rightarrow CE$$

$$C \rightarrow \epsilon$$

$$A \rightarrow C$$

$$B \rightarrow C$$

$\hookrightarrow$ si quito C mantengo ϵ

$$D \rightarrow A \mid B \mid ab$$

$\hookrightarrow$ si quito A o B mantengo ϵ

$$S \rightarrow bBb$$

$$\hookrightarrow S \rightarrow bBb$$

$$S \rightarrow bb$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow c \mid a$$

$$B \rightarrow c \mid b$$

$$C \rightarrow CDE \mid DE \mid CE \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

b) elim. prods unitarias

$$\left. \begin{array}{l} A \rightarrow C \\ B \rightarrow C \\ C \rightarrow E \\ D \rightarrow A \\ D \rightarrow B \end{array} \right\} \text{ prods. unitarias}$$

→ pares = $(S, S), (A, A), (A, C), (A, E), (B, B), (B, C),$
 $(B, E), (C, C), (C, E), (D, D), (D, A), (D, B),$
 $(D, C), (D, E), (E, E)$

→ prods no unitarias

$$C \rightarrow CDE \mid DE \mid CE$$

↓

$$A \rightarrow a \mid CDE \mid DE \mid CE$$
$$B \rightarrow b \mid CDE \mid DE \mid CE$$
$$C \rightarrow CDE \mid DE \mid CE$$
$$D \rightarrow ab \mid a \mid b \mid CDE \mid DE \mid CE$$
$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$
$$A \rightarrow a \mid CDE \mid DE \mid CE$$
$$B \rightarrow b \mid CDE \mid DE \mid CE$$
$$C \rightarrow CDE \mid DE \mid CE$$
$$D \rightarrow ab \mid a \mid b \mid CDE \mid DE \mid CE$$

c) elim. símbolos useless

i) no generadores = E, C
generadores = S, A, B, D

$$\left. \begin{array}{l} S \rightarrow aAa \mid aa \mid bBb \mid bb \\ A \rightarrow a \\ B \rightarrow b \\ D \rightarrow ab \mid a \mid b \end{array} \right\}$$

2) alcanzables = S, AB

$S \rightarrow aAa \mid bBb$

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow a$

$B \rightarrow b$